

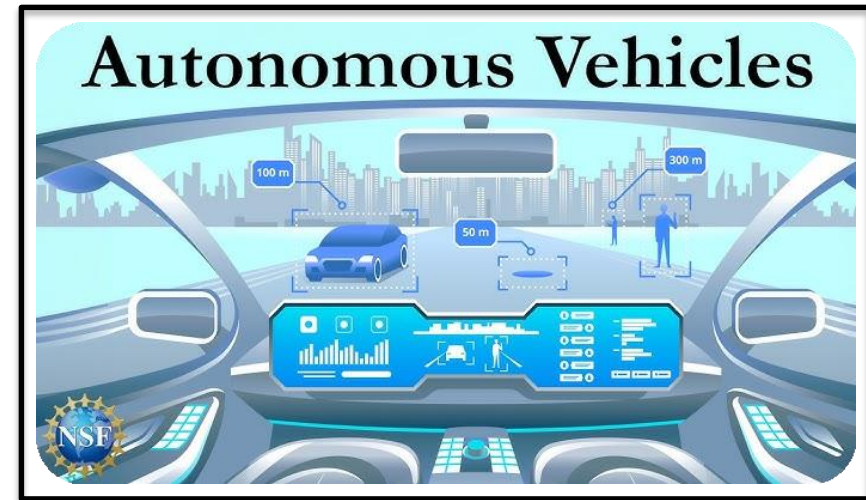
Info about the Autonomous Cars and the Aksha Project

Introduction to Autonomous Vehicles

- Autonomous vehicles (AVs) are designed to operate without human intervention, utilizing advanced technologies.

- Key benefits include reduced traffic congestion, optimized fuel efficiency, and significant decreases in accident rates due to human error.

- The development of AVs faces technical, legal, and ethical challenges that must be addressed to achieve Level 5 autonomy.



Technical Challenges in AV Development

AlgoAnalytics - An Overview

AlgoAnalytics, an NTT Data invested company, specializes in AI/ML-powered solutions with over 100+ implementations across various domains. Their expertise includes:

- Classical Machine Learning and Deep Learning
- Emerging technologies like Quantum Computing and Digital Twin
- LLM-based chatbot development
- Predictive, Image/Video, and Text Analytics
- MLOps, DevOps, FinOps, and SecOps

They are known for their proven experience in Generative AI (Gen AI) solutions, offering a comprehensive tech stack and portal for managing resources and assets from 20+ Gen AI projects.



- Sensor fusion is critical, requiring integration of LiDAR, radar, and high-resolution cameras for accurate environment perception.

- Navigating complex real-world scenarios involves understanding social cues and ambiguous situations.

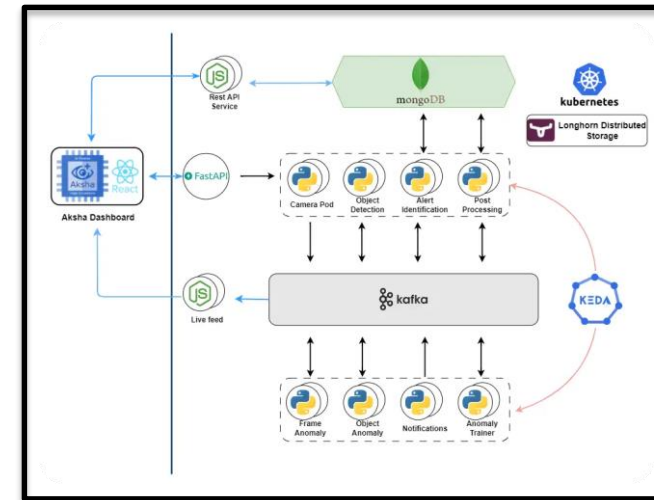
- Robust algorithms are necessary for real-time decision-making and ensuring passenger safety.

Overview of the Aksha Project

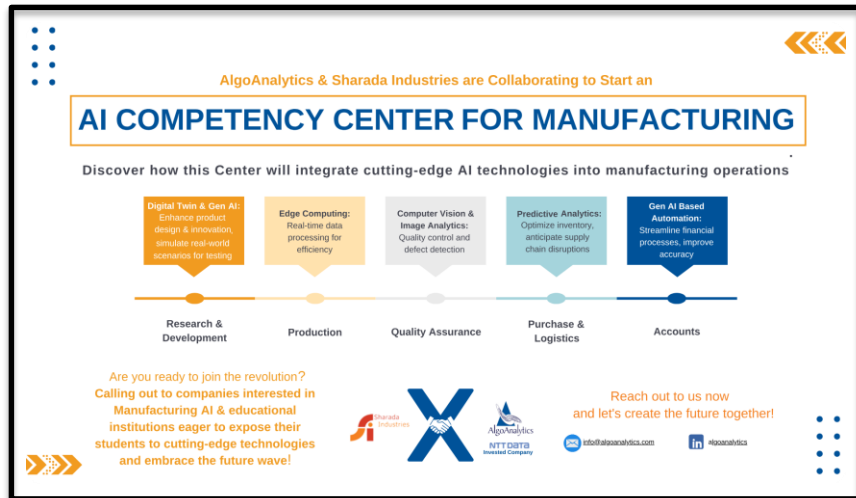
- Aksha is an intelligent surveillance system aimed at enhancing workplace safety through automated monitoring.

- Utilizes computer vision, artificial intelligence, and generative AI for real-time object detection and anomaly analysis.

- Designed to operate with minimal human intervention, providing continuous oversight of premises.



Aksha Customization and Development Phases



- Collaboration between AlgoAnalytics and NDIL led to the customization of Aksha for worker safety, running on Edge + Cloud architecture.

- Three development phases completed, with Phase III incorporating generative AI to enhance functionality.

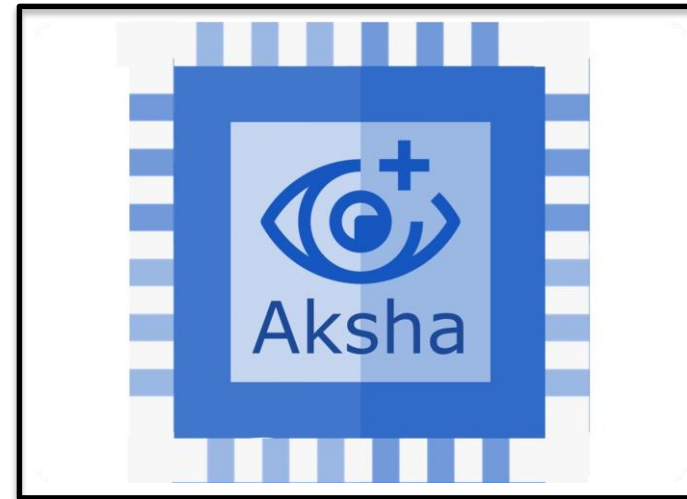
- Successful proof of concept demonstrated on Raspberry Pi 4B with wireless camera integration.

Key Features of Aksha

- Live recording and processing capabilities ensure real-time monitoring and data analysis.

- Advanced object detection includes recognition of persons, vehicles, and safety compliance (e.g., helmet detection).

- Generative AI enables unsupervised learning for anomaly detection and customized alert creation.



Business Value of Aksha

Similar projects using image/video analytics:

- **Absenteeism Prediction** for workforce for a top global **automobile company**
- **Manufacturing process monitoring** for a large company in Ahmedabad, India

- Enhances employee safety by providing immediate alerts for unsafe conditions or behaviors.

- Reduces operational costs associated with manual surveillance and incident response.

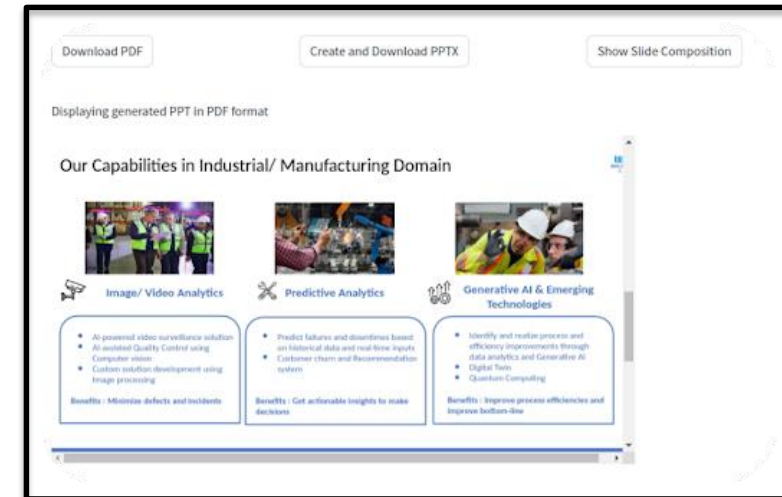
- Facilitates compliance with safety regulations through automated reporting and analysis.

Practical Applications of Aksha

- Applicable in various sectors including manufacturing, construction, and logistics for enhanced site safety.

- Can be integrated into existing security systems to augment monitoring capabilities.

- Supports data-driven decision-making through comprehensive insight report generation.



Future Directions for Aksha and AVs

- Continued development of AI capabilities to improve accuracy and responsiveness of surveillance systems.
- Potential integration of Aksha technology into autonomous vehicles for enhanced safety and monitoring.
- Exploration of ethical implications and regulatory frameworks to support the deployment of AVs and intelligent surveillance.

Contact Information and Further Resources

- For more information on Aksha, visit:
<https://aksha.algoanalytics.com/>

- Contact us at:
aksha@algoanalytics.com for inquiries and collaboration opportunities.

- Visit our office at Alacrity India Innovation Centre, Pune for direct engagement and demonstrations.

